EXACT REPAIR TRADE-OFFS FOR STRUCTURED OBSERVATION FAILURES IN A THREE-DIMENSIONAL MAJORITY RULE

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Abstract. On a finite periodic three-dimensional lattice, we study one synchronous majority update with threshold 14 on each 27-site neighborhood, including its center. We fix an observation mask for a reference configuration with two flat phases. The mask certifies the update if every completion of the unobserved bits gives the prescribed output at every site. For each admissible finite nonempty index set J, observations fail on translates of a fixed five-site tile along one coordinate direction. A repair may add or remove observed sites only within Chebyshev distance two of the failures. Failed sites remain unavailable, and all observations have their reference values.

Let $k = |J|$, let h count successive index differences equal to two, and let c count clusters formed by joining successive indices at distance at most two. Relative to the pre-failure mask, a additions and d deletions give a net increase of $\Delta = a - d$ observations and an edit count of $e = a + d$. We prove that the complete Pareto frontier consists of the $h + 1$ cost triples

$$(\Delta, e, a) = (4k + h + c - t,\ 14k + h + c + t,\ 9k + h + c), \qquad t = 0, \ldots, h.$$

Integer budget caps are jointly feasible exactly when at least one of these triples lies within all three caps. We bound acquisitions by counting which overlapping majority neighborhoods can use the same newly observed site. Given the tile positions in J, we construct a repair attaining each triple. Thus the three counts (k, h, c) suffice to decide feasibility.

1. Introduction

A set of observed bits can determine the next majority update at every lattice site even when many bits remain unknown. If some observations become unavailable, which sites should be observed to recover that guarantee? We consider a fixed observation mask for a configuration with two flat phases on a finite three-dimensional torus. Each site takes the majority of its 27-site neighborhood in one synchronous step. A repair changes the set of observed sites within Chebyshev distance two of the failures, keeps the failed sites unavailable, and must certify the prescribed output for every completion of the remaining unknown bits. The repair changes only the observation mask. Retained and newly observed bits take their values from the reference configuration.

Consider two failed five-site tiles at indices $J = \{0, 2\}$, with the intervening tile unaffected. Relative to the pre-failure mask, the ten failed observations count as deletions in every repair. One repair adds 20 observations and makes no further deletion, giving costs $(\Delta, e, a) = (10, 30, 20)$ in the order net increase, edits, and additions. A second repair also drops one previously available observation, giving $(9, 31, 20)$. Could a repair satisfy both $\Delta \le 9$ and $e \le 30$? Those caps would give $2a = \Delta + e \le 39$, whereas the acquisition bound for this pattern gives $a \ge 20$ and hence $2a \ge 40$.

In general, let J be a finite nonempty set of indices for the prescribed five-site tiles, normalized and ordered as $0 = j_1 < \cdots < j_k$ under the lattice-size and boundary assumptions of Section 2. For $1 \le r < k$, put $g_r = j_{r+1} - j_r$ and define

$$h = |\{r : g_r = 2\}|, \qquad c = 1 + |\{r : g_r \ge 3\}|.$$

Thus c counts clusters joined across gaps of one or two. Both counts use only the $k-1$ successive gaps in this list.

Theorem 2 gives the following test for integer upper bounds K, E, A on net increase, edits, and additions, respectively. A repair exists exactly when

$$K \ge 4k + c, \qquad E \ge 14k + h + c, \qquad K + E \ge 2(9k + h + c), \qquad A \ge 9k + h + c.$$

Section 3 bounds the net increase and constructs a repair. Section 4 counts how many overlapping output neighborhoods can use the same new observation. The resulting acquisition bound, together with $\Delta + e = 2a$ for a additions, gives the coupled constraint on $K + E$. By varying which old observations the construction retains, Section 5 realizes every point on the Pareto frontier.

For a fixed failure pattern, the observation requirements give a binary covering model. Florios and Mavrotas apply AUGMECON2 to enumerate the exact Pareto set of a fixed bi-objective 0–1 set-covering instance. Here the four inequalities decide feasibility for every admissible pattern from the three counts alone. The tile positions enter when we construct a repair on the original lattice. [1, §§2.2–3.1]

2. Model

Lattice and update. Let $\Lambda = \mathbb{Z}_{6m} \times \mathbb{Z}_{L_y} \times \mathbb{Z}_{L_z}$, where $m \ge 4$ is an integer, L_y, L_z are positive multiples of three, and $L_y \ge 9$. All site coordinates are read modulo the corresponding side length. Using representatives $0 \le x < 6m$, set $\rho(x, y, z) = 1$ for $x < 3m$ and $\rho(x, y, z) = 0$ otherwise. For a binary configuration f, one synchronous update gives

$$M(f)(v) = \mathbf{1}\left\{ \sum_{u \in N(v)} f(u) \ge 14 \right\}, \qquad N(v) = v + \{-1, 0, 1\}^3.$$

An observation mask $C \subseteq \Lambda$ certifies ρ if $M(f) = \rho$ for every f that agrees with ρ on C.

An exact covering condition. Put $W(v) = \{u \in N(v) : \rho(u) = \rho(v)\}$. Then C certifies ρ if and only if

$$|C \cap W(v)| \ge 14 \qquad \text{for every } v \in \Lambda.$$

Fourteen observed bits equal to $\rho(v)$ force that value at v under every completion. Conversely, if the inequality fails at one site v, assign every unobserved bit in $N(v)$ the value $1 - \rho(v)$ and extend arbitrarily elsewhere subject to the observations. At most thirteen bits in $N(v)$ then equal $\rho(v)$, so this completion fails at v.

The pre-failure mask. In each phase let $i = x \bmod 3m$ and form the list $b_0, \dots, b_{3m-1}$ by repeating $(5, 9, 0)$ exactly $m - 1$ times and appending $(5, 9, 5)$. Define

$$(x, y, z) \in C_0 \quad \Longleftrightarrow \quad 3(y \bmod 3) + (z \bmod 3) < b_i.$$

Each transverse 3×3 window contains every residue pair modulo three exactly once, so layer i contributes b_i observations. Every interior three-layer sum is 14 except the terminal triple $5 + 9 + 5 = 19$. At either phase boundary the two same-phase layers contribute 14. Thus C_0 certifies ρ.

Failures and repairs. Let $J = \{j_1, \dots, j_k\}$ be finite and nonempty, with $0 = j_1 < \cdots < j_k$ and $L_z \ge 3(j_k + 1) + 6$. With $P = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1)\}$, set

$$D = \{(3, y, 3j + z) : j \in J,\ (y, z) \in P\}, \qquad N_2(D) = \bigcup_{v \in D} (v + \{-2, -1, 0, 1, 2\}^3).$$

Since $b_3 = 5$, the five residue pairs in P have precisely the observed ranks $0, \dots, 4$. The translated tiles lie in disjoint three-column blocks within $0 \le z < L_z$. Hence $D \subseteq C_0$ and $|D| = 5k$. A repair is a mask C that certifies ρ and satisfies

$$C \cap D = \varnothing, \qquad C \triangle C_0 \subseteq N_2(D).$$

Every site outside the repaired mask is free in a completion, including each failed site and each observation voluntarily dropped by the repair.

The union $N_2(D)$ lies in layers 1 through 5 of the first phase, so the opposite phase stays fixed. Within the allowed region we impose no additional translation symmetry or fixed layer on the repair.

Relative to C_0, write $a = |C \setminus C_0|$, $d = |C_0 \setminus C|$, $\Delta = a - d$, and $e = a + d$. In particular $d \geq 5k$. With h, c as in Section 1, exactly $k - h - c$ of the successive index gaps have size one.

3. The minimum net increase in observations

Proposition. For every admissible J, the minimum of $\Delta = |C| - |C_0|$ over repairs is $4k + c$. A repair attaining this value has $a = 9k + h + c$ additions and $d = 5k + h$ deletions.

Separation. The radius-two repair region of tile j occupies columns $[3j - 2, 3j + 4]$. At a gap of at least three indices, the nearest columns of the two repair regions are at distance at least three. A majority window spans only two column steps, so it cannot use variable cells from both regions. Across the periodic seam the rightmost column of the last region is $3j_k + 4$ and the next copy of the first region starts at $L_z - 2$. Their difference is $L_z - 3j_k - 6 \geq 3$ by Section 2. Each affected inequality therefore belongs to one cluster, with all its other cells fixed to C_0. Repairs can be chosen independently and their costs add.

Translate one cluster by a multiple of three columns so that its first tile has index zero. Let p count its occupied tiles and s its gaps of two. Its indicator word $w_0 \dots w_{\ell-1}$ has length $\ell = p + s$, starts and ends in one, and has no consecutive zeros. The transverse repair region is the integer rectangle $[-2, 3] \times [-2, 3\ell + 1]$ with $(3, 3\ell + 1)$ removed. Indeed, each occupied tile covers $[3j - 2, 3j + 4]$ on rows $y = -2, \dots, 2$, and $[3j - 2, 3j + 3]$ on row $y = 3$. Successive occupied tile origins differ by at most six columns. Write S for this region. Layer 1 is forced because each layer-0 output has five fixed observations in layer 0 and needs all nine sites in layer 1. We may thus take $V = \{2, 3, 4, 5\} \times S$ as the variable set, with the failed cells fixed to zero in the mask.

Lower bound. The supplementary window table selects original output windows in layers 3 and 4, each with exactly 14 baseline observations. Let n_v count the selected windows containing $v \in V$. The table satisfies three incidence conditions.

- $n_v \in \{0, 1, 2\}$, with $n_v = 2$ on every failed cell.
- Every $v \in V \cap C_0$ has $n_v \geq 1$.
- The cells of $V \setminus C_0$ with $n_v = 2$ are exactly

$$B = \{(3, 1, 3j + 2) : 0 \leq j < \ell - 1,\ w_{j+1} = 1\}.$$

There is one member of B for each occupied tile after the first. Write $R_{\mathrm{cl}} = \{1, \dots, 5\} \times S$ for this cluster's repair region and $\delta_v = \mathbf{1}_C(v) - \mathbf{1}_{C_0}(v)$. Every changed site belongs to exactly one such region. Layer 1 is fully observed in C_0 and is forced to remain so, hence $\delta_v = 0$ on $R_{\mathrm{cl}} \setminus V$. The net increase attributable to this cluster is therefore

$$\Delta_{\mathrm{cl}} := |C \cap R_{\mathrm{cl}}| - |C_0 \cap R_{\mathrm{cl}}| = \sum_{v \in R_{\mathrm{cl}}} \delta_v = \sum_{v \in V} \delta_v.$$

Summing the selected covering inequalities gives $\sum_{v \in V} n_v \delta_v \geq 0$. Consequently,

$$\sum_{v \in V} \delta_v = \sum_{v \in V} n_v \delta_v - \sum_{v : n_v = 2} \delta_v + \sum_{v : n_v = 0} \delta_v \geq 5p - |B| = 4p + 1.$$

On each failed cell $\delta_v = -1$, so its contribution to the second term is $+1$. The only possible losses in that term occur on B, at most one per cell. Cells of multiplicity zero were unobserved and contribute nonnegatively.

Attainment. For each three-column block the supplementary repair table specifies layers 2, 3 and 5 from the tile bit and its two neighbors, using an end symbol for a missing neighbor. This construction keeps layer 4 observed and treats each end separately. An occupied block has nine additions and five forced deletions. A zero block has one addition and one deletion.

The left end adds one observation and the right end stays unchanged. This gives $a = 9p+s+1$, $d = 5p + s$, and $\Delta = 4p + 1$.

There are c clusters, with $\sum p = k$ and $\sum s = h$. Summing their costs gives $a = 9k + h + c$, $d = 5k + h$, and $\Delta = 4k + c$. Each of the h deletions at zero blocks can be reversed. Restoring any subset preserves every covering inequality. Section 5 uses this freedom to meet the joint budgets.

Verification of the local tables. In block j, an entry with column residue r refers to column $3j + r$ in both tables. The end blocks have indices -1 and ℓ, with origins -3 and 3ℓ. The tables use only the two neighboring tile bits in addition to the central bit. A window reaches at most one column to either side, so its observed support, or the multiplicity at a given cell, depends on at most five consecutive tile positions, with end markers where needed. Any such interior context can be made into a complete word by adjoining at most one occupied tile at each end. At a marked end we keep the actual boundary and extend only the open side. In both cases a word of length at most seven suffices. Translation by three columns preserves the reference mask. The size assumptions keep periodic copies of the repair region at least three columns apart in z and four rows apart in y, so a window never draws variable cells from two copies.

There are 33 complete words of lengths one through seven with occupied endpoints and no consecutive zeros. For each word the integer verifier reconstructs the radius-two region and checks the following finite assertions.

(1) The constructed mask omits all failed cells and satisfies the covering inequality at every affected output.
(2) The total numbers of additions and deletions are $9p + s + 1$ and $5p + s$. To check the block counts separately, we grouped the changed sites in the returned mask by $\lfloor z/3 \rfloor$ and counted additions and deletions in each group, including both end blocks.
(3) Every window selected for the lower bound has exactly 14 baseline observations.
(4) The multiplicities computed from all selected windows satisfy the three conditions on n_v above, including the stated set B.

The five-position argument covers every local check for arbitrary cluster length. Summing the block counts gives the total costs.

`PRIMAL_TABLE.json  DUAL_TABLE.json  verify_local_rule.py`

4. The minimum number of new observations

Theorem 1. For every admissible J, the minimum of $a = |C \setminus C_0|$ over repairs is $9k+h+c$.

Proof. Write $X = C \setminus C_0$ and let $n_1 = k - h - c$ be the number of successive index gaps of one. Select the following windows, each named by its center in the original lattice.

Window	Center	Column interval	Indices	Failed sites in the window
A_j	$(2, 0, 3j)$	$[3j-1, 3j+1]$	$j \in J$	$4 + \mathbf{1}_{\{j-1 \in J\}}$
B_j	$(4, 1, 3j+1)$	$[3j, 3j+2]$	$j \in J$	5
T_j	$(2, -1, 3j+3)$	$[3j+2, 3j+4]$	$j \in J,\ j+1 \notin J$	1

Each window U lies in the first phase and has exactly 14 baseline observations, from layers with counts $9, 0, 5$ or $5, 9, 0$. Window A_j meets four failed sites of tile j and, if $j-1 \in J$, the site $(3, 0, 3j-1)$ of the preceding tile. Window B_j meets all five sites of tile j. The only failed site in T_j is $(3, 0, 3j+2)$, since tile $j+1$ is absent. Any repair must have $|C \cap U| \geq 14$, hence

$$|X \cap U| \geq |(C_0 \setminus C) \cap U| \geq |D \cap U|.$$

This includes repairs with voluntary deletions. There are $k - n_1 = h + c$ windows of type T. The total demand is therefore

$$\sum_U |D \cap U| = (4k + n_1) + 5k + (k - n_1) = 10k.$$

The A and T windows together have pairwise disjoint column intervals. Their centers lie at distinct multiples of three, since T_j is used only when A_{j+1} is absent. The B windows also have pairwise disjoint column intervals. An intersection between the two families is confined to layer 3. There, a T window can meet a B window only in row 0, which is fully observed in C_0. An A_j window can also meet a B window in row 1. In that row, a site is initially unobserved precisely when $z \equiv 2 \pmod 3$. The only such column of A_j is $3j-1$, and the only B window containing it is B_{j-1} if $j-1 \in J$. These comparisons remain valid modulo L_y, L_z because the row and column intervals $[-2, 2]$ and $[-1, 3j_k + 4]$ each contain fewer sites than the corresponding period.

Thus the only sites that can contribute two incidences per acquisition are

$$H = \{(3, 1, 3j-1) : j \in J,\ j-1 \in J\}, \qquad |H| = n_1.$$

For each acquired site $v \in X$, let $m(v)$ count the selected windows containing it. Then $m(v) \leq 1 + \mathbf{1}_H(v)$, so

$$10k \leq \sum_U |X \cap U| = \sum_{v \in X} m(v) \leq a + |X \cap H| \leq a + n_1.$$

It follows that $a \geq 9k+h+c$. The repair in Section 3 acquires exactly this many observations. □

5. The Pareto frontier and simultaneous budgets

For an admissible J, write $L = 4k + c$, $a_0 = 9k + h + c$, and $E_0 = a_0 + 5k$. We order cost triples (Δ, e, a) coordinatewise, with smaller values preferred.

Theorem 2. The nondominated cost triples are exactly

$$P_t = (L + h - t,\ E_0 + t,\ a_0), \qquad t = 0, \dots, h. \tag{1}$$

For integer upper bounds (K, E, A) on (Δ, e, a), a repair exists if and only if

$$K \geq L, \qquad E \geq E_0, \qquad K + E \geq 2a_0, \qquad A \geq a_0. \tag{2}$$

If acquisitions are uncapped, omit the last inequality.

Proof. The repair of Section 3 makes a_0 additions, loses the $5k$ failed observations, and drops one further observation at each of the h holes. To obtain P_t, restore any $h - t$ of those further observations. Each restored site belonged to C_0 and remains available. Restoring the site with its reference value increases $|C \cap W(v)|$ by one when the site belongs to $W(v)$ and leaves it unchanged elsewhere. The resulting mask has $a = a_0$ and $d = 5k + t$, giving (1) by $\Delta = a - d$ and $e = a + d$.

For any repair, Sections 3 and 4 give $\Delta \geq L$ and $a \geq a_0$. Since $d \geq 5k$, we have $e = a + d \geq E_0$. The identity $\Delta + e = 2a$ gives $\Delta + e \geq 2a_0$. Thus all four inequalities in (2) are necessary.

Conversely, when $A \geq a_0$, the constructed triple P_t lies within the caps precisely when the integer t satisfies

$$\max(0, L + h - K) \leq t \leq \min(h, E - E_0).$$

Both endpoints are integers. Since $h \geq 0$ and $L + h + E_0 = 2a_0$, this interval is nonempty precisely when the first three inequalities in (2) hold. Choosing its left endpoint gives a repair within the caps.

It remains to show that (1) exhausts the frontier. Use the costs of any repair as its own caps. The preceding construction gives a P_t no larger in any coordinate. Conversely, suppose a feasible $Q = (\Delta', e', a')$ satisfies $Q \leq P_t$. Its third coordinate satisfies $a_0 \leq a' \leq a_0$, so

$$\Delta' + e' = 2a' = 2a_0 = (L + h - t) + (E_0 + t).$$

The first two coordinates are bounded above by the corresponding coordinates of P_t and their sums agree. Both bounds are equalities, hence $Q = P_t$. Thus each P_t is nondominated. The first coordinate decreases strictly with t, so the list has exactly $h + 1$ distinct cost triples. □

For real upper bounds, take the floor of each cap before applying (2), since all three costs are integers. As an example with integer caps (Figure 1), take $J = \{0, 2, 4\}$. Here $(k, h, c) = (3, 2, 1)$ and the frontier consists of $(15, 45, 30)$, $(14, 46, 30)$, and $(13, 47, 30)$. Caps $K = 13$, $E = 46$ allow a sum of only 59, whereas every repair needs at least 60. Increasing K to 14 makes the middle triple feasible provided $A \geq 30$.

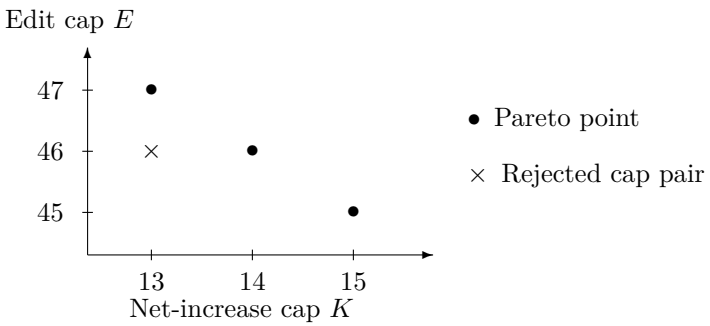


Figure 1. Repair budgets for the three failed tiles at indices $0, 2, 4$. Each dot is attained with 30 new observations. The cap pair marked by the cross allows only $13 + 46 = 59$ units in total. Every repair requires at least 60.

6. What information determines a repair budget?

Complete inputs. For an admissible input J, Theorem 2 uses only (k, h, c) to decide the budget caps. Each coordinate can affect the answer when the other two are fixed. In each row below, both instances are on the same torus, acquisitions are uncapped, and the pair (K, E) bounds net increase and edits.

Coordinate varied	Feasible J	Infeasible J	(K, E)
k	$\{0\}$	$\{0, 1\}$	$(5, 15)$
h	$\{0, 1\}$	$\{0, 2\}$	$(9, 29)$
c	$\{0, 1\}$	$\{0, 3\}$	$(9, 29)$

These examples establish that no coordinate can be dropped from this particular triple while retaining the other two.

Joining ordered inputs. For each nonempty consecutive block of the sorted input, retain its first and last indices and compute (k, h, c) from gaps internal to the block. Let U precede V and put $g = \min V - \max U > 0$. The counts for their concatenation are

$$k = k_U + k_V, \qquad h = h_U + h_V + \mathbf{1}_{\{g=2\}}, \qquad c = c_U + c_V - \mathbf{1}_{\{g\le 2\}}.$$

Its endpoints are $\min U$ and $\max V$. The join introduces exactly one successive gap. It contributes to h when $g = 2$ and merges two clusters when $g \le 2$. Since the rule returns the counts and endpoints of the concatenated list, repeated ordered joins give the same result under either parenthesization. All joins use the normalized linear order of Section 1 and count only gaps between successive indices in that order. The empty block has no endpoints. Joining it to another block returns the other block.

For example, the prefixes $\{0, 3\}$ and $\{0, 4\}$ both give $(k, h, c) = (2, 0, 2)$. Appending the same index 6 gives $(3, 0, 3)$ and $(3, 1, 2)$, respectively. At $(K, E) = (14, 46)$ the first fails because $4k + c = 15$. The second admits a repair of cost $(14, 46, 30)$. Thus the three counts alone do not determine the effect of a common continuation. Retaining the endpoints supplies the missing gap.

Producing a witness. Given the actual indices and feasible caps, choose $t = \max(0, L + h - K)$ from Section 5. Feasibility gives $t \le \min(h, E - E_0)$, so P_t meets the caps. Apply the repair table to the clusters of the full input and restore $h - t$ optional deletions. The output lists the sites added to and removed from C_0. At a join of gap one or two the neighboring tile types can change, so the construction reads the joined cluster before placing its observations. For infeasible caps, the procedure reports the violated inequalities in (2) and supplies the window and failed-site lists. The applicable lower bounds are as follows.

Violated cap	Certificate
$K < L$	The weighted count of Section 3 gives $\Delta \ge L$.
$E < E_0$	Section 4 gives $a \ge a_0$ and the $5k$ failed observations give $d \ge 5k$, hence $e \ge E_0$.
$K + E < 2a_0$	The acquisition bound gives $\Delta + e = 2a \ge 2a_0$.
$A < a_0$	The windows and shared sites of Section 4 give $a \ge a_0$.

For sorted input the counts require one scan. Across all clusters the repair tables visit $k + h \le 2k - c$ tile positions and two ends per cluster. Each table entry specifies a bounded number of sites and windows. Thus the sparse construction takes $O(k)$ table and incidence operations, counting coordinate arithmetic at unit cost. The supplied implementation also sorts its input and output lists. Checking a proposed repair uses the original coordinates, mask and neighborhoods of Section 2. `budget_oracle.py` · `check_oracle.py`

7. Verification and comparison with existing methods

Scope of the checks. Three computations play different roles in the proof.

Computation	What is checked	Role in the argument
Section 3, 33 complete tile words	The repair and window tables are evaluated on the original 27-site neighborhoods. The verifier checks coverage, costs and the multiplicities in the lower certificate.	These are finite premises of the proof. The five-position dependence and the treatment of endpoints in Section 3 extend them to arbitrary cluster lengths.
Section 4, 38 test inputs	The selected windows have baseline support 14, total loss $10k$, and exactly the predicted set of doubly counted acquisition sites.	The tests check the implementation. The coordinate count in Section 4 proves the acquisition bound for every admissible J.
Sections 5–6, 32 torus cases and 5,814 arithmetic cases	The torus checker rebuilds both window counts. For a feasible case it applies the edits to C_0, checks the allowed region and costs, and computes every output with the unknown bits first all zero and then all one. For an infeasible case it checks the reported violation. The arithmetic cases compare the integer budget test with the list of triples.	Monotonicity places every completion of each tested mask between its two endpoints. The integer-interval argument in Section 5 gives budget equivalence for arbitrary inputs and caps.

The proof for arbitrary cluster lengths uses the 33-word check together with the local-dependence argument of Section 3. The remaining suites test the implementations. The universal conclusions in Sections 4–6 follow from the window count, the integer budget argument and the ordered-gap calculation, respectively.

The three entry points are `verify_local_rule.py`, `check_acquisition_certificate.py`, and `check_oracle.py`. The last script calls the supplied repair procedure and checks its outputs on periodic arrays.

What follows from existing methods? For fixed J, the coverage constraints of Section 2 and the costs (Δ, e) give a bi-objective binary integer program. Its Pareto set can be enumerated by the ε-constraint method described by Florios and Mavrotas. The identity $a = (\Delta + e)/2$ then recovers the acquisition cost at each point. To obtain one formula for every J, Sections 3 and 4 identify the sites shared by tight windows and construct compatible repairs. Once their sharp bounds and witnesses are available, the four budget inequalities follow by eliminating the single integer t. [1, §§2.2–3.1]

A direct circular-matrix argument encounters an obstruction already for $J = \{0\}$ on the $24 \times 9 \times 9$ torus. Bianchi, Nasini, Tolomei and Torres prove integrality of fixed-integer-sum slices for covering polyhedra whose matrix has circular rows in a common column order. In the original window-by-site matrix take the rows centered at $(2,0,0), (2,0,2), (2,2,1)$ and the columns indexed by $(2,0,1), (2,1,2), (2,1,0), (2,0,4)$, in those orders. The columns are available sites in the repair region. Their incidences are

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

A circular order for the original matrix would induce one on the four displayed columns because deleting the other columns preserves each row's interval of ones. On the first three columns alone, all three pairs are adjacent around a circle. With the fourth column present, the row supports $\{1,3\}$, $\{1,2\}$ and $\{2,3\}$ cannot all be adjacent pairs in a four-cycle. The original matrix therefore fails the circularity hypothesis of the fixed-sum integrality lemma. In the proof of that lemma, total unimodularity enters after a change of variables. [2, §2 and Lemma 3.1]

Separately, the square submatrix on the first three columns has determinant 2. This excludes total unimodularity of the original covering matrix. Both conclusions concern this window-by-site formulation. A reformulation may have a different matrix structure.

8. Discussion

For every admissible arrangement of the five-site failures in Section 2, Theorem 2 determines all feasible integer upper budgets on net increase, edits and acquisitions. Every point on the Pareto frontier uses the same minimum number of new observations, $a_0 = 9k + h + c$. The tile indices locate a repair attaining any chosen frontier point, while the three counts (k, h, c) determine its cost.

Corollary (Combined budget slack). Let $L = 4k + c$ and $E_0 = 14k + h + c$ be the separate minima of net increase and edits. Suppose acquisitions are uncapped or $A \geq a_0$. For nonnegative integers u, v, the caps $K = L + u$ and $E = E_0 + v$ are jointly feasible if and only if

$$u + v \geq h.$$

Indeed, $2a_0 - L - E_0 = h$, so this is exactly the remaining inequality in (2). When $u + v = h$, choosing $t = v = h - u$ gives

$$P_t = (L + u, E_0 + v, a_0).$$

Thus every integer split of the minimum combined slack is attained, and the two separate minima are attained together exactly when $h = 0$. Each gap of two supplies one available observation that the construction in Section 3 may retain or drop. Retaining it increases net observations by one and saves one edit, with no change in acquisitions.

For a different set of failed sites, the covering condition of Section 2 still tests whether a given mask certifies the same majority update. The acquisition count in Section 4, however, uses the exact overlap of windows meeting the prescribed five-site tiles. Attainment also uses the compatible repair table of Section 3. Keeping the reference mask, update rule and repair radius fixed, for which other families of repairable failure sets D does every nondominated cost triple have acquisition coordinate $a_{\min}(D)$, the minimum possible for that D?

References

[1] K. Florios and G. Mavrotas, Generation of the exact Pareto set in multi-objective traveling salesman and set covering problems, *Applied Mathematics and Computation* **237** (2014), 1–19. Author manuscript, MPRA Paper 105074.

[2] S. Bianchi, G. Nasini, P. Tolomei and L. M. Torres, On dominating set polyhedra of circular interval graphs, arXiv:1712.07057v2 (2018).

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